LABORATORY REPORT 15-20

Part A: ASK & FSK

Q1: What feature of the ASK signal suggests that it is a form of Amplitude Modulation (AM)?

The key feature that shows that Amplitude Shift Keying (ASK) is related to Amplitude Modulation (AM) is the behavior of its signal envelope. In ASK, the envelope of the carrier signal directly follows the shape of the digital modulating signal. Mathematically, the ASK signal can be represented as the product of the message signal m(t)m(t)m(t) and the carrier signal c(t)c(t)c(t). In this process, the amplitude of the carrier is switched between two discrete levels depending on the digital input—one level representing logic "1" and another representing logic "0". This is very similar to a conventional AM signal that is fully modulated (100% modulation) by a square-wave message. Because the information is conveyed by varying the carrier amplitude, ASK is essentially a digital form of amplitude modulation.

Q2: Why must the carrier frequency be much higher than the bit rate?

The carrier frequency in ASK systems must be significantly higher than the bit rate to ensure proper separation between the carrier signal and the digital information signal. This large difference allows the receiver's Low-Pass Filter (LPF) to effectively remove the high-frequency components associated with the carrier while preserving the slower transitions of the digital data. If the carrier frequency were too close to the bit rate, the filtering process would not clearly separate the carrier ripples from the data waveform. As a result, the digital pulses would become distorted or smeared, potentially causing errors due to intersymbol interference where adjacent bits overlap in time.

Q3: In FSK, what is the relationship between the digital logic levels and the carrier period?

In Frequency Shift Keying (FSK), information is transmitted by switching the carrier frequency between two different values corresponding to binary logic levels. Since frequency and period are inversely related according to the equation T=1/fT = 1/fT=1/f, a higher frequency results in a shorter period, while a lower frequency results in a longer period. In the experiment, a logic "1" is represented by a higher carrier frequency, which produces shorter oscillation periods. Conversely,

a logic “0” is represented by a lower carrier frequency, which produces longer oscillation periods. This variation in oscillation timing allows the receiver to distinguish between the two digital states.

Q4: Why is FSK generally more robust against noise than ASK?
FSK systems tend to perform better in noisy environments because the information is encoded in the signal’s frequency rather than its amplitude. Most environmental noise sources primarily affect the amplitude of signals by adding random fluctuations. Since ASK relies on amplitude changes to represent data, these fluctuations can easily corrupt the transmitted information. In contrast, FSK receivers focus on detecting frequency changes, typically by monitoring the timing of zero-crossings or using frequency discriminators. Additionally, FSK receivers often use amplitude limiters that clip variations in signal strength, effectively removing amplitude noise while preserving the frequency information needed for data recovery.

Part B: BPSK & QPSK

Q5: What happens to the carrier at the point of a bit transition in BPSK?
In Binary Phase Shift Keying (BPSK), the information is encoded in the phase of the carrier signal. When a bit transition occurs, the carrier experiences a sudden phase change of 180 degrees. This means that the waveform effectively flips its polarity. For example, if the carrier signal is at a positive peak when the bit changes, it immediately transitions toward a negative peak, creating a phase inversion of the sine wave. This abrupt phase shift allows the receiver to detect the change in binary state while maintaining a constant amplitude and frequency for the carrier signal.

Q6: Why is an Envelope Detector ineffective for BPSK demodulation?
An envelope detector works by tracking variations in the amplitude of a signal. However, in BPSK systems the carrier amplitude remains constant regardless of whether the transmitted bit is a logic “1” or logic “0”. The only parameter that changes is the phase of the carrier waveform. Because the envelope detector only responds to amplitude variations, it would simply produce a constant DC output when applied to a BPSK signal. Since no amplitude differences exist between the two binary states, the detector would be unable to recover any useful information from the signal.

Q7: How does QPSK double the data rate without increasing bandwidth?
Quadrature Phase Shift Keying (QPSK) increases data transmission efficiency by encoding two bits of information in each transmitted symbol. It accomplishes this by using two carrier signals that are orthogonal to each other—typically sine and cosine components. These two carriers represent the in-phase (I) and quadrature (Q) channels. Each channel carries one bit simultaneously, allowing two bits to be transmitted during a single symbol interval. Because the symbol rate becomes half of the total bit rate, the required transmission bandwidth—which depends on the symbol rate—remains essentially the same as that of BPSK, while the overall data throughput is doubled.

Q8: What is the effect of a phase error in a QPSK local carrier?
In a QPSK receiver, accurate phase alignment between the transmitted signal and the local carrier reference is essential. If a phase error occurs in the local oscillator, the orthogonality between the I and Q channels is disrupted. This causes signal energy from one channel to leak into the other, a phenomenon known as cross-talk or IQ leakage. As a result, the receiver may incorrectly interpret the transmitted symbol, significantly increasing the Bit Error Rate (BER). Even small phase mismatches can degrade system performance because the decision boundaries between constellation points become distorted.

Part C: DSSS

Q9: Define "Processing Gain" in the context of DSSS.
Processing Gain in a Direct Sequence Spread Spectrum (DSSS) system refers to the ratio between the high-speed chipping rate of the pseudorandom noise (PN) sequence and the lower data bit rate of the original message signal. This parameter indicates how much the transmitted signal is spread across a wider frequency bandwidth. A higher processing gain improves the system's ability to resist interference and noise because the spreading process distributes the signal energy over a large bandwidth. During reception, the signal is despread using the same PN sequence, which compresses the desired signal back to its original bandwidth while spreading interfering signals and noise, effectively reducing their impact.

Q10: What is the "Spreading" process mathematically?
The spreading process in DSSS is performed by combining the original message signal with a pseudorandom noise (PN) sequence. In digital implementations, this is typically done using an Exclusive-OR (XOR) operation between the message bits m(t)m(t)m(t) and the PN sequence p(t)p(t)p(t). When represented in bipolar form, where logical values are mapped to ±1, this operation becomes equivalent to multiplying the message signal by the PN code. The result is a signal whose bandwidth is significantly larger than that of the original message because each data bit is replaced by multiple high-rate chips from the PN sequence.

Q11: Why must the receiver be perfectly synchronized with the transmitter's PN code?
Synchronization between the transmitter and receiver PN sequences is critical in DSSS systems because the despreading process relies on correlation. To correctly recover the original message, the receiver must generate an identical PN sequence that aligns exactly with the incoming signal. If the locally generated PN code is even slightly misaligned—such as being shifted by one chip period—the correlation between the two sequences drops dramatically. This prevents the signal from collapsing back to its narrowband form, making it nearly impossible for the receiver to extract the original message from the spread spectrum signal.

Part D: SDR & Undersampling

Q12: How does Undersampling relate to the Nyquist–Shannon sampling theorem?
Undersampling, also known as band-pass sampling, does not actually violate the Nyquist–Shannon sampling theorem but instead applies it in a different context. In traditional baseband sampling, the sampling frequency must be greater than twice the highest frequency component of the signal (fs>2fmaxf_s > 2f_{max}fs>2fmax). However, for band-pass signals that occupy a limited bandwidth BBB, the sampling requirement changes to fs>2Bf_s > 2Bfs>2B. This means that even signals with very high carrier frequencies can be sampled accurately as long as their bandwidth is relatively small and proper filtering is used. This technique allows high-frequency signals to be processed using lower-speed analog-to-digital converters (ADCs).

Q13: What is the primary role of the Band-Pass Filter (BPF) in an SDR front-end?

In a Software Defined Radio (SDR) system, the Band-Pass Filter (BPF) plays a crucial role in preparing the signal before it reaches the analog-to-digital converter. The BPF restricts the incoming signals to a specific frequency range, effectively isolating the desired signal from other unwanted signals outside that band. By limiting the input to a single Nyquist zone, the filter prevents signals from other frequency regions from folding or aliasing into the same baseband frequency after sampling. Without this filtering stage, multiple signals could overlap during the sampling process, resulting in interference that cannot be easily removed in the digital domain.

Q14: Calculate the alias frequency if a 100 kHz carrier is sampled at 9 kHz.

The alias frequency can be calculated using the formula:

$$f_{alias} = |f_c - n \cdot f_s|$$

where f_c is the carrier frequency, f_s is the sampling frequency, and n is the nearest integer multiple of the sampling frequency to the carrier frequency.

First, determine the value of n:

$$n = \frac{f_c}{f_s} = \frac{100}{9} \approx 11.11$$

The closest integer value is $n = 11$.

Substituting into the formula:

$$f_{alias} = |100 - (11 \times 9)|$$

$$f_{alias} = |100 - 99|$$

$$f_{alias} = 1 \text{ kHz}$$

Therefore, when a **100 kHz** signal is sampled at **9 kHz**, it will appear as a **1 kHz alias frequency** in the sampled signal.